

Technology Validation and Start Up Fund

Round 44 Proposal Evaluations

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TECHNOLOGY VALIDATION AND STARTUP FUND

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1) Executive Summary

Redwood is a Columbus, Ohio based LLC founded by former Battelle executives over 13 years ago. Redwood has assembled an extraordinary team for this Program. Each member of the seven-person Redwood team is an accomplished technology commercialization professional with decades of experience in performing business and technical evaluations. This team, combined with identified external subject matter experts, has extensive experience in all six of the Ohio Third Frontier technology focus areas. More detail on the Redwood team is provided in Appendix 1 of this report and on our website (<https://redwdinnov.com>). Details of the TVSF program and the review process are provided in Appendix 2.

Twenty-two (22) TVSF Round 44 applications, totaling \$4,950,000 were received: three Phase 1 applications totaling \$1,200,000 (1 Phase 1, Option 2 and 2 Phase 1 Option 3 applicants) and 19 Phase 2 applications totaling \$3,750,000. Three Phase 2 applicants were found to be non-conforming to the TVSF guidelines and were not reviewed. Three Phase 1 and 16 Phase 2 applications were reviewed, upon which 2 additional Phase 2 applications were found to be non-conforming and omitted from further consideration. The eligibility-adjusted total for this round was \$4,000,000 (\$1,200,000 Phase 1 and \$2,800,000 Phase 2).

Funding is recommended for the three Phase 1 applications totaling \$1,200,000 and seven Phase 2 applications for a total of \$1,400,000, yielding a combined total of \$2,600,000. This translates to a 59% recommended application funding rate for Phase 1/Phase 2 TVSF Round 44, compared to the average of 51% over all 44 TVSF rounds. The recommended approval rate for Phase 2 proposals in this round is 50%.

2) Evaluation Results

Evaluation summaries and funding recommendations for the Round 44 proposals are shown below in Tables 1 and 2. The total recommended funding for Phase 1 and Phase 2 projects is \$1,200,000, and \$1,400,000, respectively. Note that the Table 1 and 2 column widths are proportional to the weighting of the evaluation criteria. For example, in Table 1, Selection Committee which is weighted at 20 is four times as wide as Project Selection which is weighted at 5. Note that a yellow evaluation indicates that the proposal meets that particular criterion.

More detailed evaluations and recommendations for each Phase 1 and Phase 2 proposal may be found in Section 3 of this report.

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Table 1
Phase 1: Proposal Evaluation and Funding Recommendation
TVSF Round 44

Proposal #	Applicant	Requested Funding	Recommended Funding	Selection Committee	Deal Flow & Budget Strategy	External Participation	Track Record	Metric	Project Management & Experience	Project Selection Process
Option 2										
FY26-8348	Cincinnati Children's Hospital Medical Center	\$ 1,000,000	\$ 1,000,000							
	Sub-Total	\$ 1,000,000	\$ 1,000,000							
Option 3										
FY26-8347	Bowling Green State University	\$ 100,000	\$ 100,000							
FY26-8349	University of Akron Research Foundation	\$ 100,000	\$ 100,000							
	Sub-Total	\$ 200,000	\$ 200,000							
	Total	\$ 1,200,000	\$ 1,200,000	Column width is proportional to score weighting in each category						
Evaluation Scale				Weak	Meets	Exceeds				

Option 1: \$200,000 to \$500,000, 1:1 cost share, 1 year to allocate funds, 2 year project period.

Option 2: \$1,000,000, 1:1 cost share, 2 years to allocate funds, 3 year project period.

Option 3: \$100,000, internship component, 1 year to allocate funds, 2 year project period.

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Please note: The applicants are listed in the following table by recommended and not recommended for funding. In each section, the applicants are listed in numeric order by proposal number.

Table 1
Phase 2: Proposal Evaluation and Funding Recommendation
TVSF Round 44

Proposal Number	Lead Applicant	Requested Funding	Recommended Funding	Team	Opportunity/Market Size	IP Protection	Proof of Concept	Potential Investor/ Business Partner Engagement	Business Model	Project Plan/Budget Narrative	Growth Plan in Ohio	ESG Interaction
FY26-8351	Clinical Research Delivered	\$200,000	\$200,000									
FY26-8355	Hyphen Innovations, LLC	\$200,000	\$200,000									
FY26-8358	LittleSeed, Inc	\$200,000	\$200,000									
FY26-8362	Resilient Medical Corp	\$200,000	\$200,000									
FY26-8364	Snow Troopers	\$200,000	\$200,000									
FY26-8366	Vitruvian, LLC	\$200,000	\$200,000									
FY26-8368	Wokkah	\$200,000	\$200,000									
	Sub-Total, \$	\$1,400,000	\$1,400,000									
FY26-8350	American Farm Enterprise, Inc	\$200,000	\$0									
FY26-8352	Fairmont Medical Innovation, LLC	\$200,000	\$0									
FY26-8353	Genosera	\$200,000	\$0									
FY26-8354	Guarded Edge, LLC	\$200,000	\$0									
FY26-8359	Nexa Institute, Inc	\$200,000	\$0									
FY26-8360	Nofzinger Life Products, Inc	\$200,000	\$0									
FY26-8367	Vosmosis	\$200,000	\$0									
	Sub-Total, \$	\$1,400,000	\$0	Column width is proportional to score weighting in each category								
	Total, \$	\$2,800,000	\$1,400,000									
Evaluation Scale				Weak	Meets	Exceeds						

Table 3 lists the funding approval rate by TVSF round. This round's approval rate of 59% of the total submitted proposals is above average over all rounds. The Phase 1 applications recommended for approval comprised 18% of the total submittals for this round. The historical range of recommendations for individual rounds has spanned 27 – 100%, with an average of 51%.

TECHNOLOGY VALIDATION AND STARTUP FUND

Table 3. TVSF Approval Rate by Round

TVSF Round 44 Funding & Approval Rate by Round

Round	\$ Recommended	% Recommended
Phase 1 & Phase 2		
1 (APR 2012)	\$950,000	35%
2 (AUG 2012)	\$900,000	52%
3 (DEC 2012)	\$610,000	44%
4 (JUN 2013)	\$864,000	30%
5 (FEB 2014)	\$1,462,000	46%
6 (JUN 2014)	\$998,000	39%
7 (OCT 2014)	\$1,100,000	57%
8 (FEB 2015)	\$710,000	37%
9 (JUN 2015)	\$550,000	31%
10 (DEC 2015)	\$925,000	38%
11 (APR 2016)	\$1,239,000	46%
12 (OCT 2016)	\$3,537,269	46%
13 (MAR2017)	\$1,567,500	38%
14 (SEP 2017)	\$498,832	27%
15 (DEC 2017)	\$2,250,000	38%
16 (MAR 2018)	\$2,098,600	52%
17 (SEP 2018)	\$2,100,000	42%
18 (DEC 2018)	\$1,150,000	35%
19 (APR 2019)	\$2,250,000	43%

Round	\$ Recommended	% Recommended
Phase 1 & Phase 2		
20 (NOV 2019)	\$1,350,000	43%
21 (FEB 2020)	\$3,944,000	56%
22 (JUN 2020)	\$1,398,630	53%
23 (DEC 2020)	\$900,000	50%
24 (MAR 2021)	\$2,092,900	55%
25 (JUN 2021)	\$800,000	75%
26 (OCT 2021)	\$1,700,000	55%
27 (FEB 2022)	\$850,000	43%
28 (APR 2022)	\$2,499,976	64%
29 (JULY 2022)	\$850,000	100%
30 (OCT 2022)	\$3,700,000	71%
31 (JAN 2023)	\$100,000	50%
32 (APR 2023)	\$850,000	64%
33 (JULY 2023)	\$1,100,000	73%
34 (OCT 2023)	\$250,000	33%
35 (JAN 2024)	\$800,000	59%
36 (APR 2024)	\$2,650,000	80%
37 (JULY 2024)	\$1,798,000	82%
38 (OCT 2024)	\$6,475,000	65%
39 (JAN 2025)	\$1,400,000	50%
40 (APR 2025)	\$1,800,000	59%
41 (JUL 2025)	\$1,162,956	40%
42 (OCT 2025)	\$2,000,000	48%
43 (JAN 2026)	\$1,789,996	50%
44 (APR 2026)	\$2,600,000	59%
Total Funding	\$70,621,659	
Average/Round	\$1,605,038	51%

Please note that the following abbreviations may be used in the one-page summaries listed below.
TAM- total addressable market; SAM- Serviceable Available Market; SOM- Serviceable Obtainable Market;
TRL- Technology Readiness Level; ERDC: Engineering Research and Development Center, U.S. Army Corp of
Engineers; DEVCOM- U.S. Army Combat Capabilities Development Command.

3) Proposal Summaries

Phase 1 Proposal Summary - Recommended for Funding

Proposal 26-8347	Bowling Green State University	Amount Requested: \$100,000
Prior Phase 1 Applications: No	<i>BGSU Technology Validation & Pre-Commercialization Program</i>	Amount Recommended: \$100,000

TOTAL BUDGET: \$100,000

Institution Snapshot: BGSU has a long-standing history of research, technology development, and commercialization, and this program is designed to provide the verification and validation need to further de-risk technologies that are candidates for licensing.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Selection Committee	Six-member committee satisfies the requirements with respect to ESP participation, external advisors, mix of entrepreneurship and technology backgrounds.
Y	Deal Flow; Budget Strategy	20+ licensable technologies are available with emphasis in advanced materials, advanced manufacturing, and sensors. Anticipate funding two projects in the first year.
Y	External Participation	Validation activities for individual projects will use independent third parties. Selection Committee members may help identify vendors.
Y	Track Record	BGSU has a history of faculty invention startups, most still early stage. They encourage entrepreneurship through active recruitment, standing educational programs and special events.
Y	Metrics	Metrics to evaluate program success include licenses, product development milestones, TVSF Phase 2 applications, shortened timelines, and company fundraising, among others.
Y	Project Management & Experience	Director of the BGSU Paul J. Hooker Center for Entrepreneurial Leadership will manage the program and will co-manage projects in conjunction with the tech transfer office.
Y	Project Selection	A two-stage process is anticipated. Criteria are appropriate.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: Bowling Green State University is adding a TVSF Phase 1 program to their existing entrepreneurship programs. All the structural elements needed for the Phase 1 program are in place, including internal funding for the intern. The Selection Committee has the appropriate expertise to help guide the program as well as make individual project selections. Their plans for the first year are appropriately modest for a new TVSF Phase 1 program.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8348	Cincinnati Children's Hospital Medical Center	Amount Requested: \$1,000,000
Prior Phase 1 Applications: Yes	<i>Cincinnati Children's Innovation Ventures Acceleration Program</i>	Amount Recommended: \$1,000,000

TOTAL BUDGET: \$2,000,000

Institution Snapshot: Cincinnati Children's Innovation Ventures Acceleration Program will be used for the needed resources, expertise, and infrastructure to progress technologies with commercial potential until they are commercialization ready.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	Very strong six-member committee has significant experience and expertise in commercializing new technologies, fund raising and venture capital in relevant biomedical areas.
G	Deal Flow; Budget Strategy	The portfolio currently holds 406 active projects with a priority focus on 25 therapeutics, 7 diagnostics & medical devices and 21 digital health & care delivery projects.
G	External Participation	CCIV has an expansive network of external providers that are used as independent third parties to perform the work.
G	Track Record	Lengthy track record in TVSF Phase 1 projects with nearly 50 listed as viable pipeline candidate exits in less than 7 years and 8 in less than a year.
G	Metrics	CCIV has an extensive list of metrics tracked that include patents issued, federal and other follow-on funding, number and amount of equity investments, product sales and license income.
G	Project Management & Experience	The project management continues to evolve as needs are identified. Projects are closely monitored and technology manager, EIRs, and SMEs among others, have a role in the process.
G	Project Selection	Thorough and well-conceived triage and down selection process.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: Cincinnati Children's Hospital has a very well-established process for selecting, monitoring and exiting viable candidate technologies. They continue to improve the process as opportunities arise. The track record is exceptional and the metrics tracked are impressive. This applicant is recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8349	University of Akron Research Foundation	Amount Requested: \$100,000
Prior Phase 1 Applications: Yes	<i>Ohio Third Frontier Technology Validation and Startup Fund Round Phase 1 Round 44</i>	Amount Recommended: \$100,000

TOTAL BUDGET: \$100,000

Institution Snapshot: The University of Akron Research Foundation (UARF) Spark Fund will provide the proof needed to bring University of Akron (UA) technologies in areas of institutional strength to the point where a scalable startup can be built based on the technology.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Selection Committee	The five-member selection committee has complementary skill sets, entrepreneurial, industry knowledge and fund management/raising adequate to select deserving candidate faculty and technology opportunities.
Y	Deal Flow; Budget Strategy	In response to solicitations for Spark Fund UARF received ten (10) Project Pitches in 4-5 Projects Advanced to Due Diligence 2022, seven (7) in 2023, four (4) in 2024, and three (3) in 2025.
Y	External Participation	Funding is spent mainly on third party services, although some standard tests are run by UA for convenience and cost savings. No list of providers was shared.
G	Track Record	33 previous Phase 1 projects funded in five Spark Fund cycles, spanning 2017 and 2022–2025, resulting in 17 funded projects, 9 licenses to startups and 1 startup exit.
Y	Metrics	Metrics identified include Startups formed, Licenses signed, follow-on funding and company exits.
Y	Project Management & Experience	One UARF Senior fellow and UARF personnel are assigned to manage each project when it is funded.
Y	Project Selection	There is a clearly outlined 4 step process to select projects.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: UARF has developed a process to help projects fund critical early steps to mitigate risks to commercialization. There has been significant personnel turnover in the last 18 months. One person from UARF attended the interview. The proposal is recommended for funding.

Phase 2 Proposal Summaries - Recommended for Funding

Proposal 26-8351	Clinical Research Delivered	Amount Requested: \$200,000
<i>Licensing Institution</i>	Research Institute at Nationwide Children's Hospital	Amount Recommended: \$200,000
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	Clinical Research Delivered
Software/Information Technology		Rev1

Company Snapshot: CRD is developing a decentralized clinical trial delivery model that brings study visits directly to participants in their homes or local community settings, at times convenient for them and their families.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Expert team: CEO >20 yrs as exec healthcare & marketing. CMO/NCH inventor is former VP of Clinical & Translational Research at Nationwide Children's. COO has 35 yrs nursing and clinical trial operations. Board has experienced healthcare executives.
G	Opportunity/Market Size	SOM is \$516M, with a plan to capture 3% (\$15.4M) by Year 5.
Y	Intellectual Property Protection	In negotiations with NCH for 3 internal disclosures covering the software concept and IT operations.
Y	Proof of Concept	Product addresses customer needs and pain points. Project will result in part of the overall software package to allow real-world demonstration of the concept.
Y	Potential Investor/Business Partner Engagement	In a business relationship with main partner. Met with customers.
G	Business Model	Business model is logical, addresses pain points, and charges customers for the solutions to those pain points.
Y	Project Plan/Budget Narrative	TVSF funding will provide initial platform and sales pipeline.
G	Growth Plan in Ohio	Ohio based and leveraging Ohio resources.
G	ESP Interaction	Engaged with Rev1.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Clinical Research Delivered (CRD) addresses patient retention, a crucial need for clinical trial monitoring and a source of significant expense and time lost for company sponsors, especially in pediatric clinical trials. CRD has already engaged with a highly qualified partner for the field work. The IT product platform itself will enable self-scheduling for study participants and clinicians, will coordinate site logistics, and ensure availability of supplies. The business model provides funding from sales to fund company development and operations, minimizing the need for dilutive capital.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8355	Hyphen Innovations LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Air Force Research Laboratory	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 28,30	Hybrid Electric Turbine Engine (HETE) Enabling Extended-Range, Low-Emission Aviation
Advanced Manufacturing		The Entrepreneurs' Center

Company Snapshot: Hyphen Innovations is developing a hybrid electric turbine engine (HETE) propulsion system designed to extend range and endurance for next-generation aircraft platforms from 50-110 Miles to 250-400 miles.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The management team has technical, market application and business experience in the areas of relevance for the opportunity. Previously raised \$1.5M in seed funding for growth.
Y	Opportunity/Market Size	Overall market for AAM aircraft is ~\$10-15B. Estimate for future aeroengines may be influenced by external market forces. It is advisable to explore applications in parallel markets.
Y	Intellectual Property Protection	U.S. Patent No. 10,897,182 issued, expires 8/2036. Application submitted for partial exclusive license to cover small scale turbomachinery.
Y	Proof of Concept	Enabling technologies have TRLs of 4-6. TRL for the integrated product offering needs to be defined. Successful completion of the project will provide a TRL of 5-6 enabling future commercialization.
Y	Potential Investor/Business Partner Engagement	Immediate future revenue from Federal grants and contracts. Discussion initiated for investment of \$6.5M for commercialization.
Y	Business Model	The business model may be influenced by external market factors. Initial commercialization may be outside aeroengine applications.
Y	Project Plan/Budget Narrative	Can demonstrate capability in 12 months.
Y	Growth Plan in Ohio	Planned expansion to 60 employees and revenues of ~\$22M by Y5.
Y	ESP Interaction	Actively engaged with The Entrepreneurs Center since 2022.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Hyphen Innovations seeks to integrate performance and manufacturing technologies that will extend the range of hybrid electric turbine engine propulsion systems for eVTOL aircraft. Initial demonstrations are planned for use in the aeroengine market sector. External market forces may affect the rate of adoption for federal and commercial applications. It is recommended that alternative markets for the technology be investigated to expand the opportunities for successful commercialization and demonstration of economic viability.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8358	LittleSeed, Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	Cincinnati Children's Hospital Medical Center	Amount Recommended: \$200,000
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: 17	<i>Clinical Aromatherapy: Project OASIS</i>
Biomedical/Life Sciences		Entrepreneurs' Center

Company Snapshot: Project OASIS (Olfactory Assistance for Serenity and Integrative Support) is an initiative to advance a child-safe comfort device designed to help hospitalized children manage nausea, anxiety, and pain.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Founder/CEO with a successful track record in healthcare. Product engineer and industrial designer with relevant experience. Part-time fundraiser and a cadre of clinical and hospital ops advisors.
G	Opportunity/Market Size	TAM=\$6.8B; SAM=\$215M; SOM=\$2.3M (yr3); SOM=\$12.5M (yr5)
Y	Intellectual Property Protection	In negotiation for exclusive rights to a provisional patent covering the device. Three other internal disclosures at Cincinnati Children's Hospital.
G	Proof of Concept	The Pufflet aromatherapy system is currently at Technology Readiness Level (TRL) 6, validated through sustained clinical use at Cincinnati Children's Hospital.
G	Potential Investor/Business Partner Engagement	In discussions with several mission-aligned investors.
G	Business Model	The team understands the market drivers and mechanisms of hospital procurement in detail.
Y	Project Plan/Budget Narrative	The project will result in design-for-mfg. and market preparation.
G	Growth Plan in Ohio	Cincinnati based. Using Ohio companies future mfg., dist. and sales.
Y	ESP Interaction	Engaged with the Entrepreneurs' Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The team is experienced and has a thorough understanding of the market needs and hospital procurement systems, all of which position the company for success. The Pufflet clinical aromatherapy system was invented by nurses answering unmet clinical needs. Early prototypes have been in continuous clinical use since 2017 at Cincinnati Children's Hospital, currently in two inpatient units. LittleSeed, Inc. was created to apply engineering to patient-focused needs. It is located in the Hilliard City Lab Tech Incubator, which provides a high-tech ecosystem for Hilliard-based companies to test and prove their concepts with low risks involved. There are currently 9 start-up companies located in the incubator.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8362	Resilient Medical Corp	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio University	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 43	<i>Implantable Foam Medical Device</i>
Biomedical/Life Sciences		TechGrowth Ohio/JumpStart

Company Snapshot: The proposed technology is a materials system platform technology crucial for developing a pipeline of medical devices in the wound treatment and void occlusion device spaces.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Team is led by a capable CEO who has particular strengths in product development and regulatory. Part time team members have been added who complement the CEO.
G	Opportunity/Market Size	The overall market is in the billions of dollars. The initial target market is a foam for reconstructive surgery of the face and neck.
Y	Intellectual Property Protection	Two patents have been filed claiming low toxicity ingredients that deliver the appropriate physical properties.
Y	Proof of Concept	Current development status is TRL 5 (demonstration in a model system). The TVSF proof of concept covers the key requirement areas.
Y	Potential Investor/Business Partner Engagement	The company has been able to raise angel investment and grant funding. Discussions with equity investors are underway.
Y	Business Model	The company plans to introduce the product via KOLs at Northwestern Medical Center then sell via specialty distributors.
Y	Project Plan/Budget Narrative	The project plan is appropriate to attract investment/partners.
Y	Growth Plan in Ohio	The company states a commitment to grow in Ohio.
Y	ESP Interaction	Strong interaction with TechGrowth Ohio & JumpStart.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Resilient has made substantial progress since their Round 43 application and is recommended for funding. Notable positive changes include the business and clinical team, path to market and initial product.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8364	Snow Troopers	Amount Requested: \$200,000
<i>Licensing Institution</i>	Naval Postgraduate School	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 41, 43	<i>Snow Troopers Software Development</i>
Software/Information Technology		Entrepreneurs' Center

Company Snapshot: Snow Troopers will build a web/mobile SaaS platform to automate and improve the efficiency of contractor scheduling and routing for snow removal services using a patented multi-agent pathfinding algorithm to reduces delays and improves service coverage.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	CEO & COO management team has operational experience in snow removal business. Advisors have been added to management team with fundraising, SaaS operations expertise. Contractors have been identified for web/mobile/GPS software development expertise.
Y	Opportunity/Market Size	Software platform will be managed by applicant to improve operating efficiency of multiple contractor snow removal activities. TAM \$2.5B for Great Lakes Region/residential/small-commercial customers.
Y	Intellectual Property Protection	Applicant to license Navy Patent 11,828,023: Method and System for Multiagent Pathfinding. Proprietary customer and contractor lists and local knowledge will provide competitive advantage.
Y	Proof of Concept	Proposal states "functional MVP was exercised manually across two seasons" ... across 20+ accounts with ~120 sites involving hundreds of service tickets". Multi-agent routing algorithm requires validation.
Y	Potential Investor/Business Partner Engagement	Preliminary discussions with non-equity grant and loan agencies and private investors. Projected capital investment need is \$2.0M.
Y	Business Model	Revenue from on-demand/subscription services plus project sharing. Projected Revenue ~\$20M, ~\$15M GM, ~\$7M NM (2030).
G	Project Plan/Budget Narrative	Draft proposals received from software development vendors.
Y	Growth Plan in Ohio	Five years projected Ohio increase is 15-18 FT employees.
Y	ESP Interaction	Entrepreneurs' Center reviewed proposal and provided guidance.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Management team has been filled out with the addition of outside advisors with Start-up business, fund-raising, and SaaS operations expertise. Software development vendors have provided draft proposals and budgets to execute the proposed project. Inventor of the licensed IP algorithms has been contacted to review the licensed routing algorithm and envisioned operations management software program. Deployment of the envisioned software package applying the licensed multi-agent routing algorithm has the potential to improve snow removal services efficiencies leading to general societal benefits and higher economic growth.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8366	Vitruvian LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	U.S. Army Engineer Research and Development Center	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	HALO: Modular Outdoor Living Structure
Advanced Manufacturing		Entrepreneurs' Center

Company Snapshot: This project advances a **Modular Assembly Shelter (MASH)** system that enables rapid, small-crew assembly of commercial outdoor structures using hand tools only, eliminating cranes, field welding, and specialty subcontractors.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	The founder and Director of Operations both have relevant start-up and fund-raising experience. There is relevant technical expertise on the management team.
Y	Opportunity/Market Size	The addressable market is estimated to be >\$200M. It is highly fragmented and the company has a unique and innovative hub model which should allow it to effectively address this issue.
Y	Intellectual Property Protection	The core technology is protected by US 10,612,233 issued April 7, 2020. A freedom to operate analysis has been performed, and licensing and CRADA discussions are underway.
Y	Proof of Concept	Current technology level is 4-5. Two current customers are ready to pilot the technology as soon as it is ready. Clear performance criteria have been established.
Y	Potential Investor/Business Partner Engagement	A potential investor is not required based on the pro forma. This will be a new product in an adjacent market.
Y	Business Model	Financial model is realistic and conservative. The company will leverage current sales force to support the new product introduction.
Y	Project Plan/Budget Narrative	Budget used to validate, produce, install a live demonstration unit.
Y	Growth Plan in Ohio	Company based in Athens and will use an Ohio supply chain.
Y	ESP Interaction	The company engaged with the Entrepreneurs Center since 2025.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The leadership team has a solid track record, the necessary experience to succeed with the MASH system, and have a solid business model. The budget will be used to provide a working unit. In addition, they provide workforce training in the region which is positively impacting the local workforce readiness.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8368	Wokkah	Amount Requested: \$200,000
<i>Licensing Institution</i>	Naval Surface Warfare Center-Dahlgren Division	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 42	<i>Secure Product Launch Platform (WokkahAI)</i>
Software/Information Technology		Entrepreneurs' Center

Company Snapshot: WokkahAI is a software execution platform designed to help founder-led teams turn ideas into working software in a more organized and reliable way.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The founder has solid technical and entrepreneurial experience. Since applying for TVSF funding in round 42, several members have been added to the team who have start-up and project management experience.
Y	Opportunity/Market Size	This is a very large potential market and the problem the company is trying to solve is real. They have good customer engagement (8 pilot customers) and a good plan to reach this fragmented customer base.
Y	Intellectual Property Protection	Plan to license Patent No. 10,762,211 from the U.S. Navy. The patent will provide a key feature to the software being developed.
G	Proof of Concept	The minimum viable product is planned for completion early in TVSF Phase 2. Very good customer engagement and input.
G	Potential Investor/Business Partner Engagement	Discussions ongoing with potential investors. Current offer for 6% of the company in return for \$150K being considered.
Y	Business Model	Business model is a subscription-based SaaS model. The pro forma financials are reasonable. The market is highly fragmented and large.
Y	Project Plan/Budget Narrative	Completed MVP at end of TVSF Phase 2.
G	Growth Plan in Ohio	Company based in OH, will add staff and use contract labor to grow.
G	ESP Interaction	They have a long-term relationship with the Entrepreneurs' Center
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The company has worked hard to address the shortcomings detailed in their Round 42 application. The team has grown and added start-up business and project management expertise. The business model has been clarified, and they have a clear vision of what their target customers look like and how to reach them.

Proposal Summaries - Phase 2 Not Recommended for Funding

Proposal 26-8350	American Farm Enterprise, Inc	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Micro farm Community Food Development System</i>
Software/Information Technology		Rev1

Company Snapshot: The Micro farm Community Food Development System is a turnkey, technology-enabled platform that turns small-scale and urban agriculture into profitable, scalable enterprises.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The three-person team is comprised of the inventor and founder, a CEO with commercialization, fundraising, and operations experience and a person with expertise in micro farm production, cooperative practices and operations, and market-facing execution.
Y	Opportunity/Market Size	The model relies on coordination of micro farms across low income, low access census tracts to provide food to municipalities and anchor nonprofits, school districts, etc.
R	Intellectual Property Protection	The IP is inconsistent with TVSF category of Software/Information Technology. Two copyrighted documents provide a template for developing micro farms and training individuals to become farmers.
Y	Proof of Concept	Currently stated as a TRL 7. The system has been piloted in Mansfield and Marion OH.
Y	Potential Investor/Business Partner Engagement	AFE states that they are having active conversations with impact investors and community development financial institutions.
R	Business Model	Sales identified to start in 2028 at 10 systems sold. Proforma and Economic impact on Ohio do not complement each other.
Y	Project Plan/Budget Narrative	Project plan actively derisks customer and investor requirements.
Y	Growth Plan in Ohio	AFE support micro farms in Ohio to add ag jobs. AFE will add FTEs.
Y	ESP Interaction	American Farm Enterprise has been meeting with Rev1.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: A very interesting application involving community engagement to develop micro farms into a coordinated local supply system capable of providing produce to local food economies. The proposal targets recruiting people to learn about micro farming techniques in a series of training exercises to teach farming techniques, business and crop plans, financial management, and farm preparation and management. The evaluation criteria for TVSF proposals includes the degree to which the intellectual property is protected relative to both the technology and the proposed business model. To be considered for the TVSF program in the Software and Information Technology area, the company will need to develop software or information technology or refine the same for successful application.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26- 8352	Fairmount Medical Innovation LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Cleveland Clinic Foundation	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	A/Z Airways
Biomedical/Life Sciences		JumpStart

Company Snapshot: Fairmont is developing a next-generation airway management platform designed to solve a recurring clinical challenge: safely transitioning between airway modalities as patient conditions evolve during procedures.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	While the team possesses foundational experience in many areas, it appears to lack experience raising capital in the MedTech space.
R	Opportunity/Market Size	It is unclear in which specific use cases this product would be used, especially given that it would likely be sold at an increased price. The beachhead market for this product is not stated.
Y	Intellectual Property Protection	The proposed product is protected by 2 issued patents, and a freedom to operate analysis was conducted by qualified counsel.
R	Proof of Concept	How the device has been tested to date, and the results of those tests, were not included in the proposal.
Y	Potential Investor/Business Partner Engagement	The team has spoken with angel and institutional investors, who have shown interest in the project.
R	Business Model	The business model is overly complex and not likely to attract investor interest.
R	Project Plan/Budget Narrative	It is unclear how the device will be refined and tested.
Y	Growth Plan in Ohio	Fairmount Medical Innovation is an Ohio-based company.
Y	ESP Interaction	The company is engaged with JumpStart.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: While the proposed product offers advantages over current solutions, the specific use case remains unclear, especially given that it will likely be sold at a higher price than current products. Given the likely niche use case for this device, the market size is unlikely to attract investment capital. Additionally, the business model and exit assumptions do not appear to be in alignment with investor and market expectations. The proposal lacks detail on how the current prototype has been tested, what refinements are needed, and if the budget is based on vendor quotes or assumptions. This application is not recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8353	Genosera, Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	Research Institute at Nationwide Children's Hospital	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 33	<i>Translational Development of GA-002: Gene Therapy for Lysosomal Acid Lipase Deficiency</i>
Biomedical/Life Sciences		Rev1

Company Snapshot: Genosera is a preclinical company developing a novel gene therapy for the treatment of Lysosomal Acid Lipase Deficiency (LAL-D), a rare genetic disease. The therapy utilizes an Adeno associated virus (AAV); the same technology used successfully for other gene therapies from Children's.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	On paper, the team is experienced in gene therapy and expert in pharmaceutical development and venture funding. However, they have not adapted to circumstances and have not shown ability to deliver.
G	Opportunity/Market Size	Primary target market exceeds \$6.8B, taking in account only those LAL-D patients in US, EP, Japan and Israel with ability to pay.
Y	Intellectual Property Protection	Genosera has obtained a license to 3 LAL-D patent applications.
R	Proof of Concept	Based on progress toward an MVP for GNE, the Round 33 funded project, the ability of the team to deliver an LAL-D MVP within a year is in doubt.
R	Potential Investor/Business Partner Engagement	Investor discussions have been put off because of lack of risk reduction and poor public relations in the AAV gene therapy space.
R	Business Model	Without a clear path to venture money, patient advocacy and/or angel investors, the future of the therapy and company are at risk.
Y	Project Plan/Budget Narrative	Plan and deliverables clearly stated. Budgets are reasonable.
G	Growth Plan in Ohio	HQ in Columbus. Fits well with Ohio technology ecosystem.
G	ESP Interaction	Long-standing relationship with Rev1 and Rev1 Ventures.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Genosera was funded in Round 33 for development of a gene therapy for GNE myopathy (GNEM). They propose to develop a second gene therapy, this one for Lysosomal Acid Lipase Deficiency (LAL-D). Progress on GNEM was severely hampered by market dynamics, some of which are likely to affect LAL-D development. The team has not presented a path to fund the company's future in a way that adapts to current market conditions.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8354	Guarded Edge LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Cleveland State University	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	HIPAA Compliance Comprehensive Management System TVSF 2
Software/Information Technology		Entrepreneurs' Center

Company Snapshot: Guarded Edge addresses HIPAA compliance by embedding required compliance activities into assignable workflows, tracked remediation, and persistent documentation, enabling organizations to execute and demonstrate compliance consistently over time.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	While the team has extensive HIPAA compliance experience, technology and fundraising expertise is limited to vendors and advisors.
Y	Opportunity/Market Size	Market sizing appears appropriate, and pricing is based on customers' current spend.
R	Intellectual Property Protection	There does not appear to be a protection strategy for a key differentiating feature from competition in the market.
Y	Proof of Concept	An MVP based on the current framework and customer feedback will be developed and tested within the project period.
Y	Potential Investor/Business Partner Engagement	The company has had preliminary conversations with local investors.
Y	Business Model	Product pricing and go to market strategy support a reasonable business model.
Y	Project Plan/Budget Narrative	Project plan appears appropriate and aligns with budget.
Y	Growth Plan in Ohio	Ohio-based company plans to attract capital and create local jobs.
Y	ESP Interaction	The company is working with the Entrepreneur's Center.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The applicant proposes a strong operational approach to HIPAA compliance; however key risks prevent funding currently. Although experienced in HIPAA compliance, the core team lacks demonstrated technology product development and fundraising experience; future investors will likely want to see these skill sets on the team prior to funding to reduce the risk for scaling and product-led growth. Also, while the lack of a clear market leader amongst competition supports the idea that their features are not sufficient to solve the customer's needs, there does not appear to be a protection strategy for the key differentiating features which will likely be required to raise additional capital.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8359	Nexa Institute, INC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Research Institute at Nationwide Children's Hospital	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 34, Digital Story Therapies	<i>Storilly: Platform for Parent-Led Speech Therapy</i>
Software/Information Technology		Rev1

Company Snapshot: Nexa Institute is commercializing Storilly, an AI-powered platform that empowers parents as primary language coaches for children with speech and language delays.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Team has complementary and appropriate technical, business, medical, specialist skills and expertise on the team.
G	Opportunity/Market Size	One in six children experience communication delays. The pediatric market is \$5.7B annually at a \$48/month subscription. The service obtainable market is \$2.7B annually.
R	Intellectual Property Protection	Patent application was assigned to Digital Story Therapies, an applicant that was funded in round 34. No new IP from NCH is included in this application.
G	Proof of Concept	A current platform MVP with a TRL of 6 has been achieved. Storilly is currently enrolled in an IRB-approved usability and prospective longitudinal research study at Nationwide Children's Hospital.
G	Potential Investor/Business Partner Engagement	Series A investors have identified three critical proof points and Storilly has raised 26% of a \$2M Friends and Family raise for 2026.
Y	Business Model	Business model with clear revenue streams identified. Appropriate business partner potential customer engagement.
Y	Project Plan/Budget Narrative	A corrected Plan/Narrative was resubmitted after the interview.
G	Growth Plan in Ohio	Current team of 8 will grow to 15-20 by 2027.
Y	ESP Interaction	Interaction with Rev1.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The co-founders of Nexa Institute and Digital Story Therapies are the same. The inventor of the Intellectual Property is also the same for both organizations. The Intellectual Property is the same. The goal of the Ohio Third Frontier Technology Validation and Start-up Fund (TVSF) is to create greater economic growth in Ohio based on start-up companies that commercialize technologies developed by Ohio institutions of higher education and other Ohio not-for-profit research institutions. This application is not recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8360	Nofzinger Life Products Inc	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Pittsburgh	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>SleepCare</i>
Biomedical/Life Sciences		Entrepreneurs' Center

Company Snapshot: The company uses patented forehead cooling to reverse overactive brain activity allowing individuals to get to sleep faster, stay asleep longer and maintain deeper sleep across the night.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	The team has entrepreneurial, technical, and marketing experience, and would benefit from capital formation and consumer medical device expertise.
R	Opportunity/Market Size	The TAM for each of the proposed products remains unclear.
R	Intellectual Property Protection	The limited remaining patent life for the proposed product is unlikely to attract investor interest.
Y	Proof of Concept	While the core device has demonstrated clinical efficacy, how the app will address specific market needs remains unclear.
Y	Potential Investor/Business Partner Engagement	The proposal states that the company is actively talking to multiple investors.
R	Business Model	The business model, go-to-market strategy and financing plan are overly optimistic.
R	Project Plan/Budget Narrative	The project deliverables remain unclear.
Y	Growth Plan in Ohio	This is an Ohio-based company and a key vendor is in Ohio.
Y	ESP Interaction	The company has engaged with the Entrepreneurs' Center.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: While the proposed product has been found to be clinically effective, the limited remaining patent coverage (2027 expiration) is unlikely to attract the investor capital needed to bring a consumer product to market. Additionally, market interest in the accompanying app, its differentiating features, and how it incorporates into the business model remains unclear.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-8367	Vosmosis	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Curriculum-aligned VR Speaking Practice for Japanese 101/102</i>
Software/Information Technology		Rev1

Company Snapshot: Vosmosis is developing a curriculum-aligned virtual reality (VR) courseware platform that extends speaking practice beyond the classroom by placing learners in realistic, repeatable scenarios (e.g., cafe ordering, transit navigation, campus conversations) and guiding them through instructor-approved tasks and dialogues.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Founders are PhD candidates with technical expertise. In addition, the Business Lead has appropriate entrepreneurial business operations, fundraising, commercialization and partnership development experience.
R	Opportunity/Market Size	Global language learning market estimated at ~\$80B. No specific TAM, SAM or SOM given for the initial market, estimates based on the number of institutions/courses/students.
Y	Intellectual Property Protection	Platform software technology licensed from The Ohio State University executed on December 14, 2025 to cover education/training applications.
R	Proof of Concept	Prototype of TRL 6 claimed, no market or external user-specified metrics identified.
Y	Potential Investor/Business Partner Engagement	Some early-stage discussions in process. Equity investments with a total of \$13M planned in Y1-4.
R	Business Model	\$15M revenue in Y5. Revenue goals from recurring SaaS license/student access fees will require rapid rate of market adoption.
R	Project Plan/Budget Narrative	Budget needs clarification to meet TVSF guidelines for contractor \$s.
R	Growth Plan in Ohio	Projected revenue of ~\$15M in Y5, no estimate of # of employees.
Y	ESP Interaction	Actively engaged with Rev1.
	Evaluator Recommendation	This application is not recommended for funding.

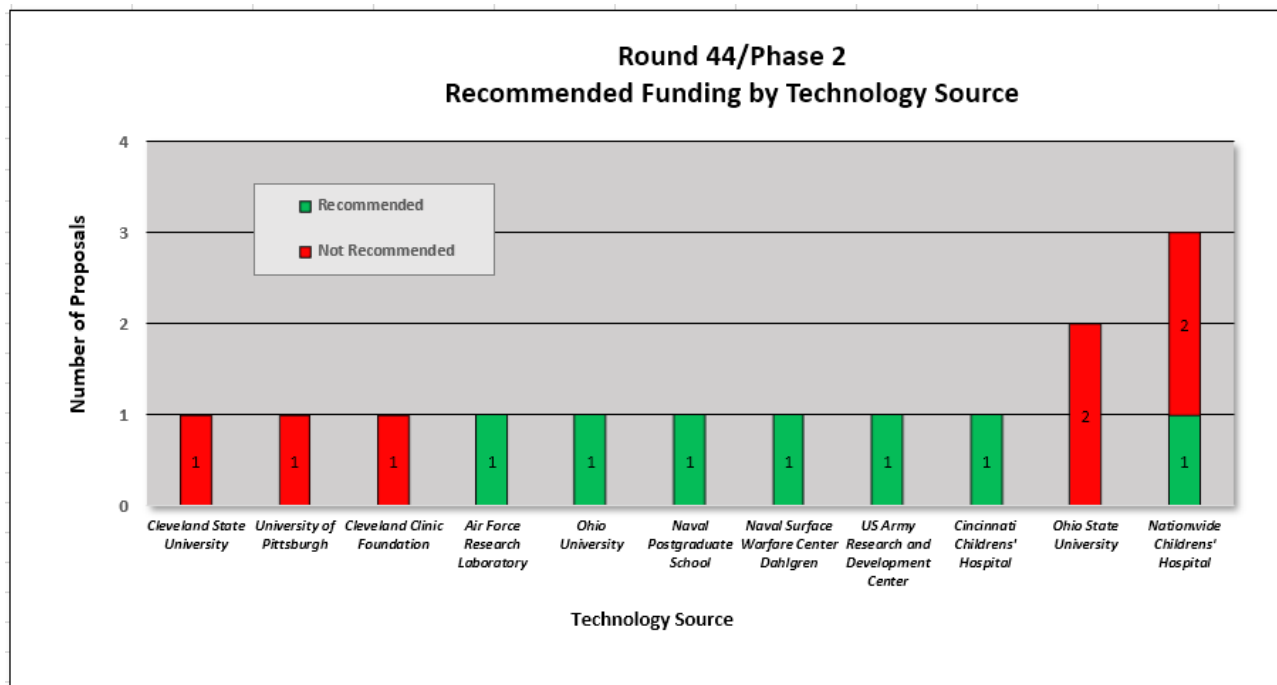
Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Vosmosis is developing a virtual reality platform that provides language training in virtual reality created social environments in which user engagement is monitored to ensure culturally authentic interactions. Quantification of the benefits and development status of integration of traditional and VR approaches would provide clear differentiation from available competitor products and support focused market adoption. A clear path for obtaining a timely and successful market entry would help focus actions that can provide the projected company growth and achieve revenue projections.

4) Round 44 Analysis

Figure 1 shows the proposal activity and funding recommendations by technology source for Phase 2 proposals. There were 11 sources of technology this round. Nationwide Children's Hospital provided three technologies; Ohio State University provided two. One of those five proposals is recommended. Single application submissions were received sourcing technology from Cleveland State University, University of Pittsburgh, Cleveland Clinic Foundation, Air Force Research Laboratory, Ohio University, Naval Postgraduate School, Naval Surface Warfare Center Dahlgren, US Army Research and Development Center, and Cincinnati Children's Hospital. One each Air Force Research Laboratory, Ohio University, Naval Postgraduate School, Naval Surface Warfare Center Dahlgren, US Army Research and Development Center, and Cincinnati Children's Hospital are recommended for funding.

Figure 1. Round Funding by Technology Source

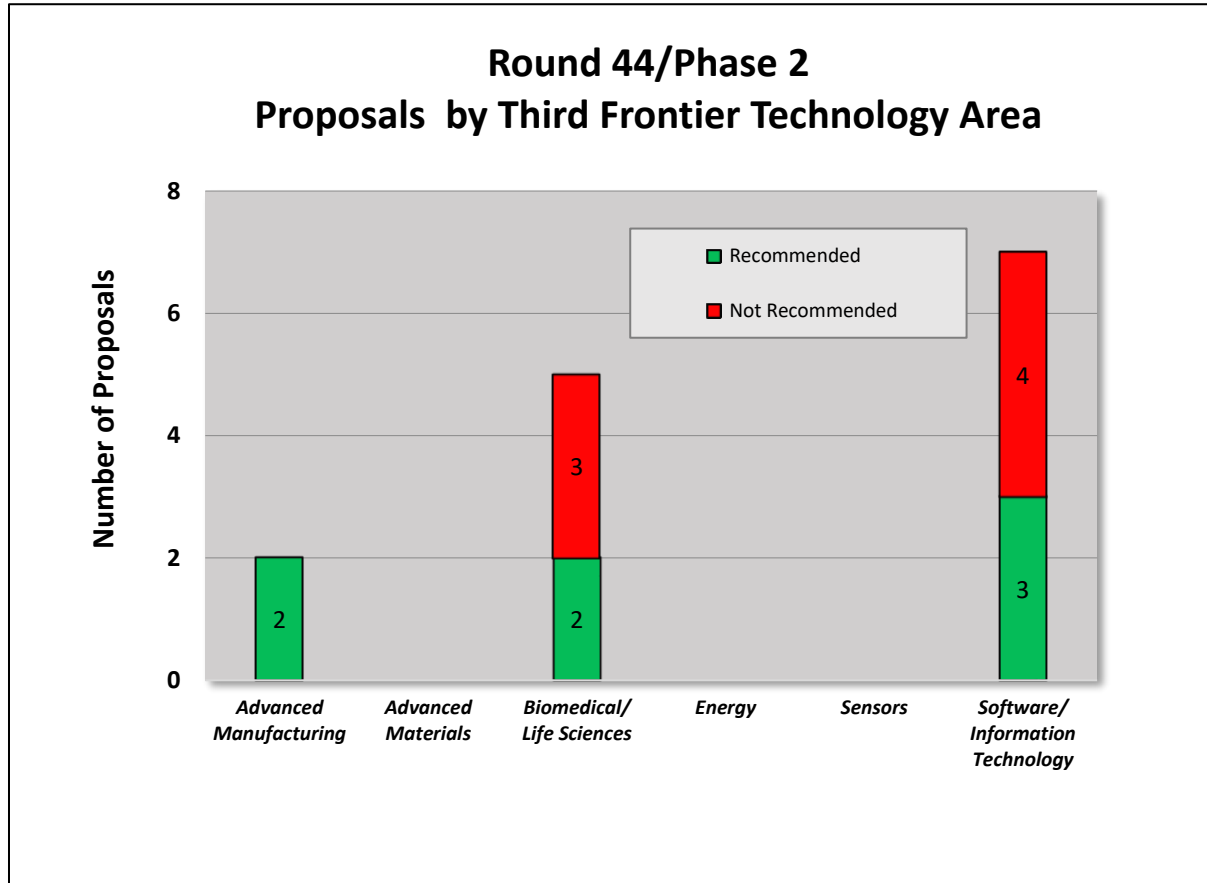


TECHNOLOGY VALIDATION AND STARTUP FUND

Figure 2 depicts Phase 2 proposal activity and funding recommendations by Third Frontier focus area. In this Round, seven of fourteen were in Software/Information Technology (50%), five of fourteen were in Biomedical/Life Sciences (36%), and one of the fourteen was in Advanced Manufacturing (7%).

Three of the fourteen in Software/Information Technology (21%), two of the fourteen in each of Advanced Manufacturing (14%) and Biomedical/Life Sciences (14%) were recommended for funding.

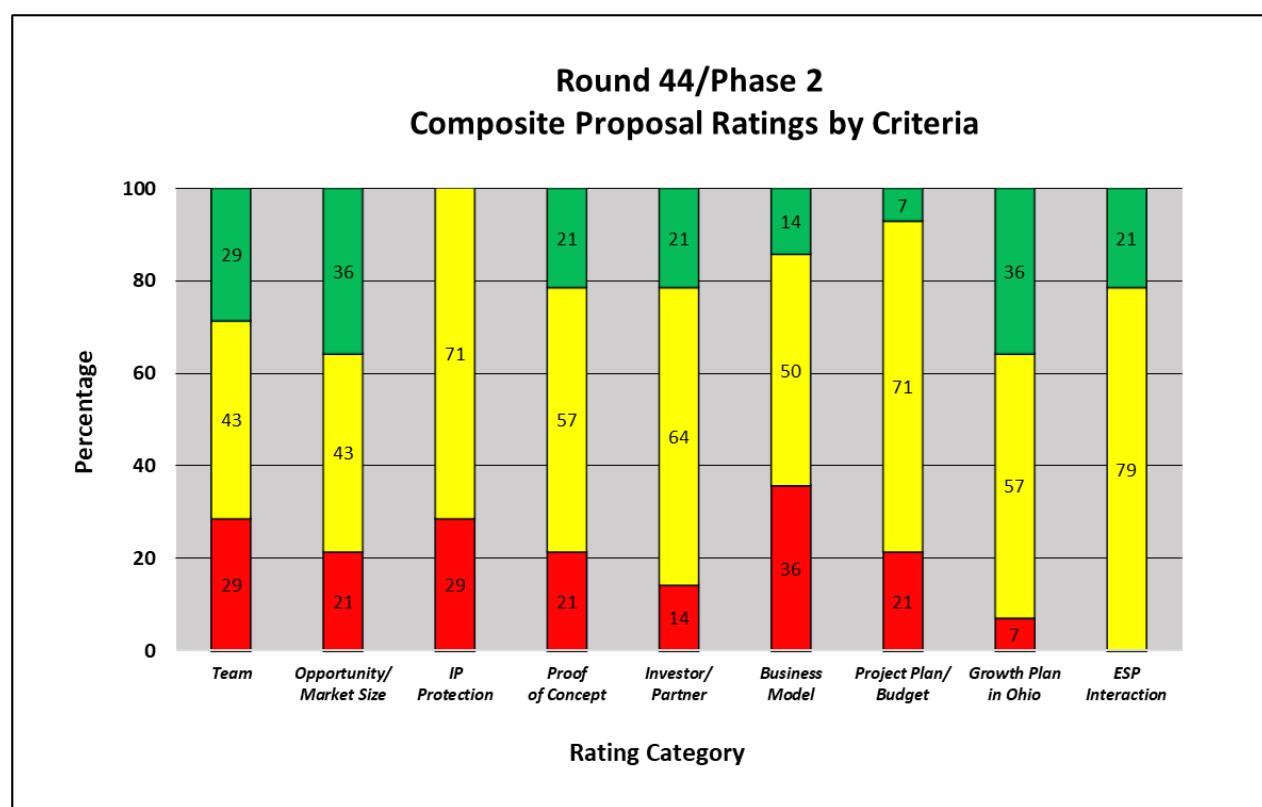
Figure 2. Round Phase 2 Proposal Activity by Third Frontier Technology Area



TECHNOLOGY VALIDATION AND STARTUP FUND

Figure 3 shows the aggregate ratings by evaluation criteria for all Phase 2 proposals. Discounting ESP interaction and Growth Plan in Ohio, Investor/Business Partner had the lowest percentage of weak (red) scores in this round. Business Model followed by Team and Intellectual Property Protection had the highest percentage of weak scores in this round.

Figure 3. Round Phase 2 Proposal Rating Summary



Of the previous twenty-four Rounds, business model was the lowest rating in Rounds 20-23 (53% average $\geq$ meets) and Round 26 (28%). The RFP was revised prior to Round 24 to elicit stronger business models and it appears that the proposals have provided stronger business models. Rounds 20 to 44 the overall average is 64%. Round 26 appears to be an anomaly. Future rounds will be monitored to see if the improvement continues.

Figure 4: Rounds 20 to 44 Phase 2 Analysis of Business Model

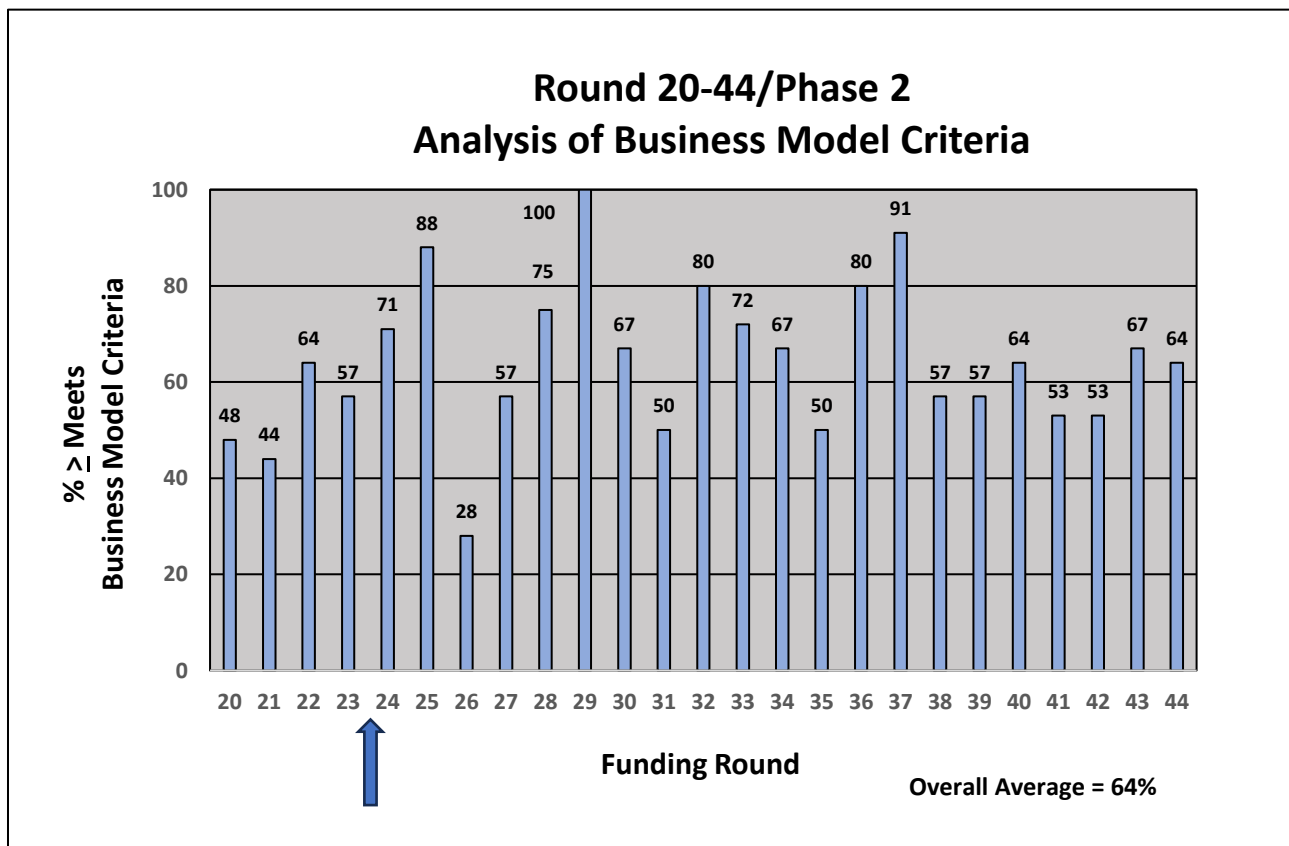
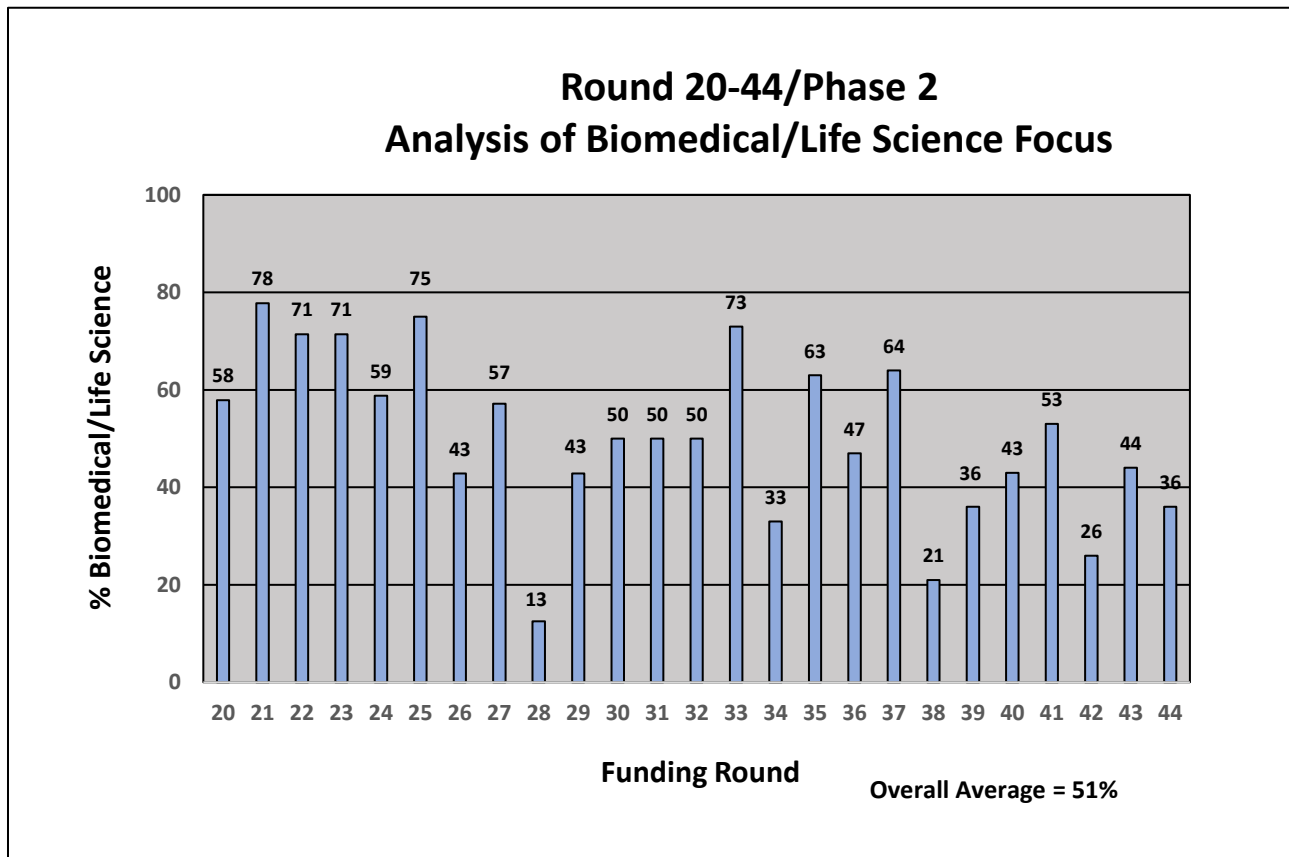


Figure 5 shows the percentage of Biomedical/Life Sciences applications for the last twenty-five Rounds. Biomedical/Life Sciences was the focus area for 36% of Round 44 applications. The overall average is 51% of the applications in Biomedical/Life Sciences. Several Software/Information technology applications in recent rounds have had Biomedical/Life Sciences focus.

Figure 5: Rounds 20-44 Phase 2 Analysis of Biomedical/Life Science Focus



Carry Through and Reapplication

Phase 1 Carry Through: There are 2 Phase 2 applicants that previously received Phase 1 funding. Both of those applications are recommended for funding.

Phase 2: There are 6 Phase 2 reapplications in this Round. Four are first time reapplications and three of the four are recommended. There are two Phase 2 second time reapplications and both are recommended for funding.

Appendix I

Summary of Redwood team and qualifications

Redwood, as a company, has been providing technology commercialization services for over 13 years while each team member has been active in this field for over 25 years.

Each Redwood team member:

- possesses an advanced technical degree and extensive business proficiency
- has worked across the spectrum of technology commercialization from invention to successful market introduction
- understands how to assess a concept case from the perspective of aligning technologies to product applications in specific markets
- has lived, both conceptually and literally, the iterative process of understanding market needs and wants, value chains and who the customers are within the value chain

Team members have all worked for major corporations, research institutions, venture capital firms and technology start-up companies gaining a comprehensive understanding of what is necessary for development teams to successfully commercialize a technology. The Redwood team has served as evaluators for the Ohio Advanced Manufacturing program and an individual team member served as an evaluator for CALF, TIP and IOF loan programs for over a decade.

The seven members of the Redwood team are highly qualified evaluators for the TVSF program and have combined experience and expertise in the following areas (combined years):

Commercializing technology into market pulled products (125+ years)

New product and new business development (240+ years)

Market/Technology Assessment (140+ years)

Startup/Spin out companies (120+ years)

Board member/Advisor to Startups (25+ years)

Evaluating/monitoring RFPs/Funding selection (150+ years)

The following is a brief summary of the seven principal team members used in this evaluation Round.

Joe Berger

- BS, University of Virginia, Chemical Engineering, MBA, Harvard University, Marketing & Finance
- Former Senior Vice President, Commercial Business-Battelle
- Former Vice President, Health & Consumer Products Business Unit-Battelle
- Former Vice President for two semiconductor equipment manufacturers
- Member of Executive Management Team (V.P. Sales) of Semiconductor Equipment Manufacturer (CFM Technologies) during IPO (1996)
- Worked directly for companies in specialty chemicals, food ingredients, semiconductor capital equipment and industrial equipment markets
- Served customers in the following industries: water treatment, pharmaceuticals, electric power generation, prepared foods, semiconductor and flat panel display manufacturing, automotive, aerospace and medical devices

Herb Bresler

- BS Biological Sciences, University of Maryland; BS Secondary Science Education, University of Maryland; PhD Immunology and Infectious Diseases, The Johns Hopkins University School of Hygiene and Public Health
- Former Senior Research Leader and Chief Scientist for Health and Life Sciences, Battelle Memorial Institute, responsible for evaluation of new technology-based business opportunities, intellectual property development, licensing and tech transfer; created and implemented new metrics to increase returns on discretionary R&D; cultivated approximately 1150 invention disclosures, 900 patent applications, and 120 granted patents, leading to \$52 million company funding
- Recipient of four R&D 100 awards for breakthrough medical devices in neuroscience and diagnostics
- Former Director of the Laboratory of Cellular Immunotherapeutics at the Arthur G. James Cancer Hospital and Research Institute at The Ohio State University

Jeff Crompton

- M.A., D.Phil., University of Oxford, Materials Science
- Founder and Former CEO AltaSim Technologies LLC
- Former Market and Technology Leader, EWI
- Former Technology Manager, Calgon Specialty Chemicals
- Former Principal Scientist, Alcan International
- Led development of innovative technology for computational analysis, medical products, aerospace and automotive markets
- Led commercial implementation of technology in production operations in North America, Europe and Asia
- Former Wolfson Industrial Fellow, University of Oxford

John McArdle

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- BE, Manhattan College, MS, Northeastern University, Chemical Engineering
- MBA, Finance/International Business, University of Chicago (Booth School of Business)
- Former Business Development Manager, Battelle
- Former Product Line Manager – Koch Industries
- Former Technical Sales Manager, Allied Signal Corporation
- Recognized expert in water and wastewater treatment technologies
- Successful track record of introducing innovative technologies for a variety of municipal, industrial, and military applications in domestic and overseas markets.

Sara Risch

- BS Biological Sciences, Kent State University; PhD Biomedical Science, The Ohio State University College of Medicine
- Former Managing Director, Rev1 Ventures
- Former Technology Mining Specialist, MTRAC Life Sciences Program (University of Michigan)
- Over 11 years evaluating academic life science innovations for commercial viability
- 5 years supporting early spinouts in developing capital access strategies to reach go/no-go milestones
- Pre-clinical spinout operator

Jim Sonnett

- BS, University of Virginia, MS, University of Massachusetts, PhD, University of Delaware, all in chemical engineering
- Former Vice President – Science and Technology, Battelle Health & Life Sciences
- Former R&D Leader – W. L. Gore & Associates and E. I. DuPont
- Built and led high impact innovation organizations in aerospace, electronics, and life sciences
- Former Board Member – Velocys, Ventaira, Battelle Ventures
- Adjunct professor – Ohio State University Fisher School of Business

Susan Stanton

- BS, Millersville University, Chemistry, MPh, Syracuse University, Organic Chemistry, PhD, University of Rochester, Organic Chemistry
- Personally developed 12+ products and led new product development teams at Mobay, Alcoa & Nexicor
- Holder of 10+ patents
- Former VP Market and Technology Assessment at the National Technology Transfer Center
- Over 15 years as an angel investor in technology-based startups
- Over 19 years as an evaluator for Ohio Third Frontier funds including IOF, CALF, TIP, and TVSF
- Over 8 years teaching market and business analytics to STEM graduate and post doc students.

Appendix 2

TVSF objectives and phases

The Technology Validation and Start-up Fund (TVSF) provides grants under two phases to transition technology from Ohio Eligible Research Institutions into the marketplace through Ohio start-up companies. Under Phase 1, Ohio Research Institutions may apply for a pool of funds to support validation/proof that will directly impact and enhance both the commercial viability of their unlicensed technologies and ability to support a start-up company. Under Phase 2, Ohio start-up and young companies may apply for funding to commercialize a technology they intend to license from a university or an Ohio research institution.

The goals of Phase 1 include:

- Generate the proof needed to move technologies to the point that they are either ready to be licensed by an Ohio start-up company or deemed unfeasible for commercialization. The institutions are encouraged to work with potential Ohio licensees to identify the proof needed.
- Perform validation activities such as demonstration and assessment of critical failure points in subsequent development, prototyping, scale-up and commercialization in order to generate this proof with strong preference for these activities being performed by an independent third-party source.

The goals of Phase 2 include:

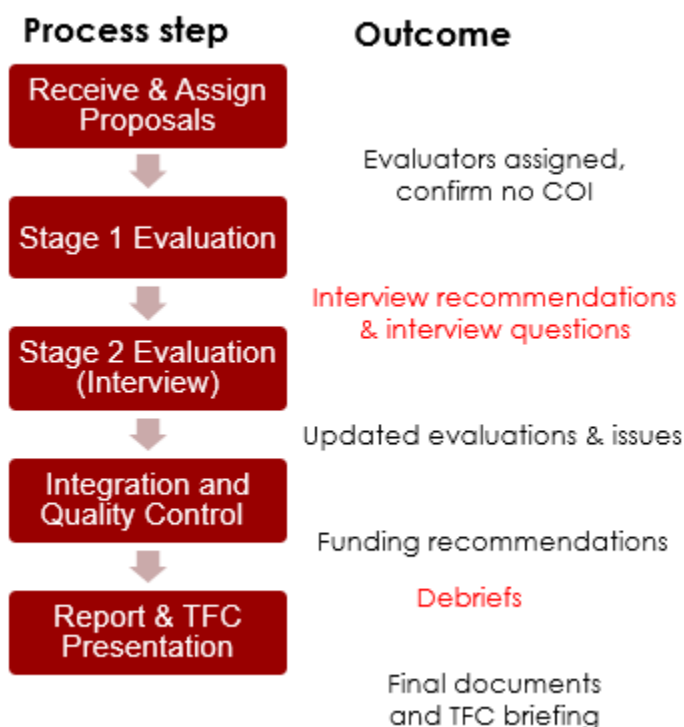
- Accelerate the commercialization of technology by Ohio start-up companies that license technology developed at Eligible Institutions during the critical early stage of the company.
- Generate the proof needed to move technology to the point where it is able to be commercialized or additional funds for commercialization can be raised. A clearly identified path to subsequent funding opportunities and working directly with potential investors to define the proof needed for investment into the company is strongly encouraged.
- Funded activities may include, but may not be limited to, beta prototype development and deployment to potential customers for testing and evaluation, and market research/business development in order to generate the proof needed.

Based upon these goals, the proposal evaluation criteria were developed. The proposals were then evaluated based on the criteria.

Description of review process

Review summary. Our overall review process flow and outcomes by stage are shown in Figure 1. A similar process has been successfully used by Redwood in prior projects for public and private clients. Discussions were held with the TVSF program manager after all but the initial step in Figure 1.

Figure 1. TVSF Evaluation Process



Review and Assign Proposal In this first step proposals were summarized and a primary evaluator was assigned who has the appropriate background and no conflict of interest.

Stage 1 Evaluation Stage 1 evaluations were conducted for each proposal using the criteria shown below in Tables 1 and 2. Differentially weighted criteria were used to evaluate Phase 1 and Phase 2 proposals. Each proposal was rated on a 0 (absent) – 5 (Outstanding) scale for each criterion, an approach used by the NSF and in other State of Ohio programs. The weightings reflect the experience of the Redwood team and our belief that some factors, for example team and market opportunity in Phase 2, are more important than others.

The entire review team subsequently discussed all the evaluations to ensure consistency and agreed upon which applicants to invite for interviews. Interview questions were then provided in advance to each applicant.

Stage 2 Evaluations (Interviews) The standard procedure for this step is: In-person, 45-minute interviews were held with each invited applicant to discuss the advance questions and other topics of interest to the

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evaluators. Two Redwood team members participated in each interview video conference, with additional team members joining, as needed.

Integration and Quality Control Proposal evaluations were updated based on interview results. A calibration review was held by the review team to ensure that evaluations were performed consistently and that any changes made were a result of team consensus. Based on this review, proposals were recommended for funding.

Table 1 – Phase 1 Evaluation Criteria

Criterion	Weighting	Description
Alignment and Compliance	Go/No go	Institutional alignment with TVSF intent and compliance with RFP
Project Selection Committee	20	Skills, background and commitment of the committee members
Deal Flow; Budget Strategy	15	Is the projected deal flow consistent with the requested budget to enable committing funds within 1 year?
External Participation	15	Does process ensure validation activities will be performed by 3 rd parties; ESPs and state-funded programs/organizations are enlisted to enhance commercialization activities of the project?
Track Record	15	Is there a strong Phase 1 or comparable program track record of licensing and newco creation? If not, is there a plan for improvement?
Metrics	15	Realism and impact of proposed metrics, including licensing, start-ups.
Project Management & Experience	15	Is there a strong project management strategy and appropriate experience of people who allocate the pool of funds and manage individual projects?
Project Selection Process	5	Is there a clear, appropriate process for project selection?

Table 2 – Phase 2 Evaluation Criteria

TECHNOLOGY VALIDATION AND STARTUP FUND

Criterion	Weighting	Description
Alignment & compliance	Go/No Go	Proposal alignment with TVSF intent and compliance with RFP
Management Team	20	Skills, background and commitment
Opportunity/market size	15	What is the market segment and total addressable market? Is it a platform or breakthrough technology or incremental improvement? If breakthrough, is it compatible with viable commercialization pathways?
IP Protection/License	15	Is IP adequately protected, does it enable the business model, is it differentiated from likely competition, is license likely within 9 months?
Compelling Proof of Concept	15	Was meaningful input from potential customers and key performance metrics used to design Proof of Concept? Are the competitive advantages compelling for potential customers?
Potential Investor/Business Partner Engagement	10	Is there company engagement/collaboration independent of licensing institution, including financial backing?
Business Model	10	Is the business model realistic AND achievable? Can the service/manufacturing model be scaled?
Project Plan/Budget Narrative	5	Is the budget consistent with proof in 1 year?
Start-up in Ohio	5	Does a start-up exist or is it planned? Will the start-up be in Ohio?
ESP Interaction	5	Is team engaged with ESP? Has team incorporated feedback from ESP into the project, proposal or business plan?